Communication-Efficient Federated Learning With Dataset Condensation for Vision Tasks

Jun Ma, *Member, IEEE*, Qi Kang, *Senior Member, IEEE*, Kefan Wang, *Student Member, IEEE*, Ruijing Sun, *Student Member, IEEE*, and MengChu Zhou, *Fellow, IEEE*

***Abstract*—Internet of Things (IoT) devices are widely distributed with large and scattered data, risk of privacy leakage and high bandwidth cost make data transmission and aggregation impractical. Federated learning (FL) is an important privacy-preserving multiparty learning paradigm, involving collaborative learning with others and local updating based on private data. Data heterogeneity is one of the main challenges in FL scenarios, which can greatly limit FL's applicability and performance. Most existing approaches tackle the heterogeneity challenge during local training or model aggregation while ignoring the performance drop caused by direct model aggregation and catastrophic forgetting in the global model. This work proposes a novel FL approach, called FL with dataset condensation (FedDC). It aims to relieve the issue of knowledge discrepancy among local models and catastrophic forgetting in the global model. Specifically, this work proposes a differential distribution matching method to summarize local data for each client without compromising on data privacy. It introduces collaborative knowledge distillation to encourage local models to learn from others. To mitigate inter-round forgetting, this work stabilizes the averaged global model's training by leveraging auxiliary information from its immediately past one. Extensive experiments are conducted. The results show that FedDC significantly outperforms the state-of-the-art methods on various vision tasks.**

***Index Terms*—Catastrophic forgetting, classification, distribution matching (DM), federated learning (FL), knowledge discrepancy.**

I. Introduction

CONVOLUTIONAL neural network (CNN) and its variants have achieved great success in a wide range of computer vision tasks [1], [2], [3], which requires centralized data collection and training that may be unacceptable in realistic Internet of Things (IoT) scenarios due to the high dispersion of IoT devices. In response to this urgent problem, the concept of federated learning (FL) has been proposed for building secure and intelligent IoT systems. FL [4], [5], [6], [7], served as a distributed learning paradigm, allows multiple clients to collaboratively train a global model without exchanging their private data. The operation of local models' updating and communicating contains much less private information than raw data sharing. Recently, FL has attracted many research interests and shown great potential to facilitate real-world IoT-based applications in many fields, including medical imaging [8], [9], object detection [10], and landmark classification [11].

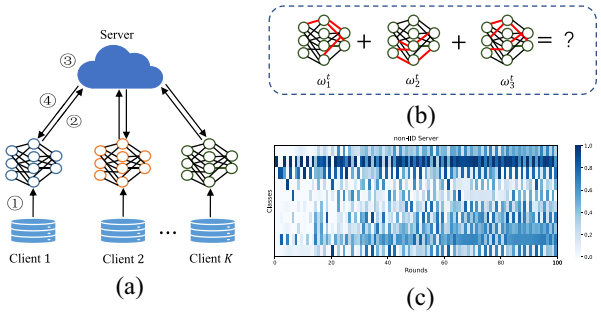


Fig. 1. (a) Problem illustration of FL. (b) Given non-independent and identically distributed (non-IID) data across clients, how can the model learn a generalizable representation? and (c) how can we effectively alleviate catastrophic forgetting to ensure the stable and satisfactory performance of the global model?

A vanilla FL algorithm is FedAvg [12]. As illustrated in Fig. 1(a), in each round of FedAvg, 1) a number of randomly selected clients are initialized with a global model and compute the model updates with its private data, 2) the updated local models are sent to the central server, 3) the server aggregates the local models' weights to obtain a new global model, and 4) the server distributes the global model to each client to conduct a new local model for the latter's update. This process is iteratively executed for many rounds until the global model converges. However, it has been confirmed that FedAvg [12] suffers from slow convergence and poor performance [13]. One explanation for this is that each client performs stochastic gradient descent (SGD) to different directions due to the inconsistent of raw data distributions [13]. Simple model aggregation drives the averaged global model away from the globally optimal one. This phenomenon is referred to as model drift. some approaches, e.g., FedProx [14] and SCAFFOLD [15] try to constrain the direction of local model update like FedProx [14] and SCAFFOLD [15]. Yet they fail to address the issues of 1) data heterogeneity [seen in Fig. 1(b)], where each client optimizes at different directions

Received 7 June 2025; revised 11 August 2025 and 1 September 2025; accepted 2 September 2025. Date of publication 15 September 2025; date of current version 20 November 2025. This work was supported in part by the Natural Science Foundation of Shanghai under Grant 23ZR1466000; in part by Tongxiang Institute of Artificial General Intelligence under Grant TAGI2-A-2024-0006; and in part by the Shanghai Industrial Collaborative Science and Technology Innovation Project under Grant 2021-cyxt2-kj10. *(Corresponding authors: Qi Kang, and MengChu Zhou.)*

Jun Ma, Qi Kang, Kefan Wang, and Ruijing Sun are with the Department of Control Science and Engineering, and the State Key Laboratory of Autonomous Intelligent Unmanned Systems, Tongji University, Shanghai 201804, China (e-mail: majun_@tongji.edu.cn; qkang@tongji.edu.cn; kefanwang@tongji.edu.cn; 2331887@tongji.edu.cn).

MengChu Zhou is with the ECE Department, New Jersey Institute of Technology, Newark, NJ 07102 USA (e-mail: zhou@njit.edu).

Digital Object Identifier 10.1109/JIOT.2025.3606800

2327-4662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

and directly averaging such updated model parameters gets away from global optimum and 2) inter-round forgetting in the global model, where the global model's performances drop dramatically after being optimized for a new round.

For the first issue, intuitively, we expect that the global model would present similar logits for the same classes in intradomains. As the clients are not allowed to reach the data owned by others, it is hard to build a shared dataset that covers all clients' data distributions. Dataset condensation (DC) [16], [17], [18], [19], [20], [21], [22] technology has emerged as a means to reduce cost related to storage, time, and computing power. Its use has the potential to overcome the limitations of privacy preservation mentioned above in FL scenarios. On the one hand, it is safe and easy to obtain compressed and diverse data. On the other hand, we can gain a more global view of all the local data distribution landscape.

Since each client introduces significantly different data distributions. when the global model is optimized with new classes from the next round of selected clients, it tends to overwrite the parameters useful for previous distributions. As shown in Fig. 1(c), the test accuracy on some classes which originally predicted well by the previous global model often drops significantly after being optimized for a new round. This implies that forgetting occurs in FL. Knowledge distillation (KD) [23], [24], [25], [26], [27], [28], [29] has been proposed to enable the local models to mimic the global model's behavior. It has been employed in FL scenarios [30], [31], [32], [33], [34]. These mentioned studies [28], [30], [35], [36], [37] have motivated us to take advantage of condensed data for knowledge transfer.

In this work, we propose a novel approach named FL with DC (FedDC) to address the issues of data heterogeneity among intradomains and catastrophic forgetting in the global model. It consists of three steps: 1) differential distribution matching (DDM) that performs DC in a data-decentralized setting; 2) collaborated KD (CKD) that ensures a positive knowledge transfer among local models; and 3) Learn with Past (LwP) [36] in the server, which provides an assisted KD mechanism to update the global model. The contributions of this work are threefold.

1) It proposes a simple-yet-efficient FL framework to tackle the challenge of data heterogeneity and knowledge forgetting, thereby effectively circumventing client drift issues.
2) Based on the collected condensed data, it employs the ensemble logits of local models as the teacher to collaboratively transfer knowledge among the local models, thus improving the generalization performance of all the local models. It enhances the averaged global model by mitigating the inter-round forgetting issue.
3) Extensive experiments on various settings are conducted and the results verify that FedDC achieves superior performance over the state-of-the-art (SOTA) methods.

Section II presents related work. Section III describes the proposed FedDC. Section IV shows the experimental settings and results. Section V offers the concluding and remarks.

II. Related Work

A. Federated Learning

FL methods enables participants to collaboratively learn a better global model by only sharing their model parameters instead of raw data [5], [12], [23], [38], [39]. In each round of FL, each client trains a local model on its own data and is required to upload its model parameters to a central server, which is used for model aggregation or distillation. However, simply averaging model parameters to update its maintained global model suffers from client drift. To copy with this problem, many studies [14], [15], [39], [40], [41] try to address the non-IID data issue. They can be divided into two categories: improvement on local training on the client side (i.e., Step 1 of Fig. 1) and on model aggregation on the server side (i.e., Step 3 of Fig. 1).

To improve local training, FedProx [14] adds a proximal term into the objective to limit the direction of local model update, while SCAFFOLD [15] introduces control variates during local training. MOON [39] constrains the local updates by maximizing the agreement of representation learned by the current local model and the representation learned by the global one. FedDyn [41] adds a linear and quadratic penalty term, whose minima is consistent with the global stationary point. Lee et al. [40] demonstrated that local training induces the knowledge outside of the local distribution on the client side. Hence, FedNTD [40] is proposed to perform local-side distillation to avoid such forgetting only for the not-true classes. To better model aggregation, FedGen [32] and FedFTG [28] explore the input space of local models through a generator, and use it to deeply transfer the knowledge from local models to the global model in a data-free manner. FedMA [38] utilizes Bayesian nonparametric methods to match and average weights in a layer-wise manner. FedAvgM [42] applies momentum on the server side.

B. Dataset Condensation

It is necessary to alleviate the challenges associated with larger datasets and train a neural network in moderate time. DC [43] has emerged to compress the larger datasets by synthesizing a compact set of informative images, and it has attracted significant attention with the explosive growth of the size of training data. Typically, DC condenses the original dataset by minimizing the distance between real and synthetic images via a predefined metric. On this basis, M3D [16] is proposed to minimize the maximum mean discrepancy (MMD) of feature representations between the synthetic and real images. Differentiable siamese augmentation (DSA) [17] further boosts the performance of DC by utilizing data augmentation techniques. To ensure privacy protection, multisize DC (MDC) [20] is proposed to condense multisize datasets by compressing multiple condensation processes into a single one. Regarding the evaluations to comprehensively reflect the generality and effectiveness of condensation methods, DC-BENCH [44] provides the first large-scale standardized benchmark on DC. Once combined with FL, the work [45] designs a differential privacy (DP)-processing method to generate a set of synthetic images by mixing DC and DP

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

techniques. DCFL [46] focuses on condensed data-assisted client model training with Non-IID awareness. The work [47] proposes a meta knowledge-driven FL paradigm by conducting meta knowledge condensation from local private data. Different from them, our work is devoted to DC to obtain the original data distribution, which is convenient for the knowledge alignment and transfer on the server side. On this basis, the current global model can better learn from clients and the past one, thus obtaining a high-performance and fast-converging global model.

C. Knowledge Distillation in FL

KD, first described in a seminal paper [48] on model compression, serves as a teacher-student paradigm to train the small model ("student"), on a transfer set with soft targets provided by the large model ("teacher"). More recently, due to its strong applicability and generality, KD has been increasingly employed in FL algorithms. For example, with the help of generated synthetic data, FedKFD [23] synthesizes consensus-labeled proxy data and use it to filter for relevant local model knowledge and train a robustglobal model through KD. FedGK [49] enables a client to effectively assimilate key knowledge from teachers and peers while minimizing extraneous peer information in FL scenarios. To generate the proxy dataset for backward distillation, Fed2KD+ [25] uses forward and backward distillation process to mitigate data and heterogeneity problems of the averaged global model by adding a cosine similarity penalty to the CVAE's training for non-IID scenarios. Though FedDF [37] maintains an auxiliary dataset that can be replaced with a pretrained generator, it does not instantiate how to acquire the generator. In summary, with the existing methods, statistical heterogeneity results in slow convergence and model drift. These methods rely on a generator or unlabeled public data to transfer knowledge from local models to the preliminary global model.

We conduct an experiment on the CIFAR10 dataset by using an unlabeled public data [37], generated data [28] and condensed data (our method). As illustrated in Fig. 3, by taking advantage of condensed data from a global view of all the local data distribution landscape, it can be guaranteed that knowledge exchanges equally among local models and further merges into the global model. Consequently, the global model is greatly enhanced.

III. Proposed Method

A. Problem Statement

Suppose that there are K parties, denoted as $P_1,\ P_2,\ \ldots,$ and P_K. Each party P_k has a local dataset $\mathcal{D}_k = \{(x_i^k,\ y_i^k)\}_{i=1}^{N_k}$ In each round, the server sends ω to a random subset of clients, and ω_k parameterizes a local model initialized by ω, each of these clients train a local model independently by performing several mini-batch SGDs on their local model

$$\ell_k(\omega_k) = \frac{1}{N_k}\sum_{i=1}^{N_k}\mathcal{L}_{\mathrm{CE}}\left(x_i^k,\ y_i^k;\omega_k\right) \tag{1}$$

where $\ell_k(\cdot)$ is the local empirical loss of client k and $\mathcal{L}_{\mathrm{CE}}$ denotes the cross-entropy loss. N_k is the corresponding number of samples.

With the help of a central server, our goal is to learn a generalized machine learning model ω over dataset $\mathcal{D} = \bigcup_{k=1}^{K}\mathcal{D}_k$, while the private raw data is not exchanged or integrated. FL can be formulated as the following problem:

$$\arg\min \omega = \sum_{k=1}^{K}\frac{|\mathcal{D}_k|}{\mathcal{D}}\ell_k(\omega_k). \tag{2}$$

In this work, each model of a client has two components: 1) a base encoder, comprising all the layers preceding the last fully connected layer in a backbone model and 2) an output layer, which denotes the global model without the last softmax layer. For ease of presentation, the base encoder $h_\omega(\cdot)$, parameterized with model weight ω, is used to extract representation vectors from inputs. Similarly, as we study the supervised setting, output layer $f_w(\cdot)$ is used to produce predicted values for all classes.

B. FedDC

In this section, we describe the proposed novel FL method: FedDC. As shown in Fig. 2, it mainly consists of three steps. First, DDM performs DC in a data-decentralized setting. Second, collaborative KD uses CKD among local models. Third, LwP provides an aggregated KD mechanism to update the global model. In the following, we present the learning objective and procedure of FedDC. The definitions of math symbols are listed in Table I.

1) Differential Distribution Matching: suppose that client k is tasked with learning a set of local condensed data denoted as S_k. The client first initializes each class of condensed data by sampling from the local original data D_k or Gaussian noise. Subsequently, the client employs a feature extractor $h_\omega(\cdot)$ to extract the feature representations from each class $c \in \{1,\ 2,\ \ldots,\ C-1\}$ of data in D_k and S_k. The means of these feature representations are computed as

$$\mu_{k,c} = \frac{1}{N_{k,c}}\sum_{j=1}^{N_{k,c}}h_\omega\left(x_{k,c}^j\right),\ \ \widetilde{\mu}_{k,c} = \frac{1}{M_{k,c}}\sum_{j=1}^{M_{k,c}}h_\omega\left(\widetilde{x}_{k,c}^j\right) \tag{3}$$

where $x_{k,c}^j \in D_k$ and $\widetilde{x}_{k,c}^j \in S_k$ represent the jth sample of class c drawn from D_k and S_k, respectively. $N_{k,c}$ refers to the number of samples of class c in D_k, whereas $M_{k,c}$ denotes the number of samples of class c in S_k. Client $k \in \{1,\ 2,\ \ldots,\ K\}$ can learn and update S_k by optimizing distribution matching (DM) [18] loss as follows:

$$\mathcal{L}_{DM}^k = \sum_{c=0}^{C-1}\left\|\mu_{k,c}(D_k) - \widetilde{\mu}_{k,c}(S_k)\right\|^2. \tag{4}$$

Other than the extraction of feature representations needed by DM loss in (4), to correctly connect DC and DP, a mild utility loss is added to the DM [18] method. To sum up, we empower client k to learn and update the local condensed data S_k by minimizing such a DDM training loss

$$\mathcal{L}_{\mathrm{DMM}}^k = \sum_{c=0}^{C-1}\left\|\mu_{k,c}(D_k) - \left(\mathcal{N}\left(0,\ \delta^2 I\right) + \widetilde{\mu}_{k,c}(S_k)\right)\right\|^2 \tag{5}$$

where $\mathcal{N}(0,\ \delta^2 I)$ is a standard Gaussian noise.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

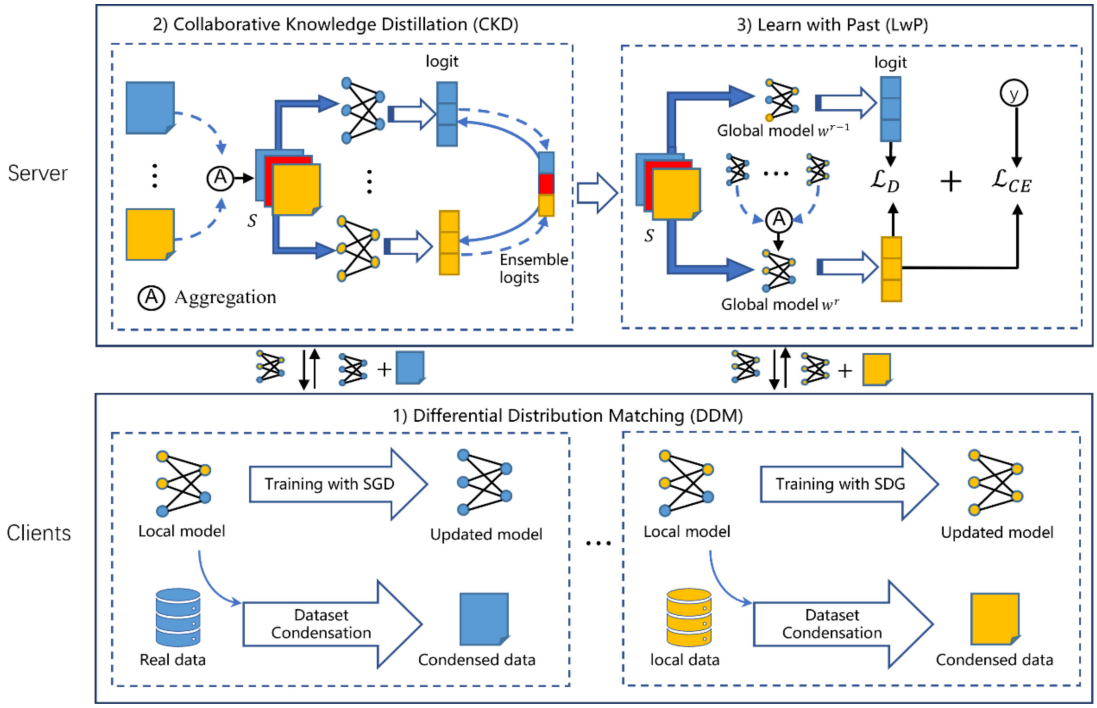


Fig. 2. Framework of FedDC that aims to address data heterogeneity and catastrophic forgetting issues in FL. 1) DDM for clients by adding a mild utility loss compared to the loss function of conventional DM method. 2) Collaboratively KD (CKD) to improve local models' generalization performance. 3) LwP to stabilize the training of global model among on the server side.

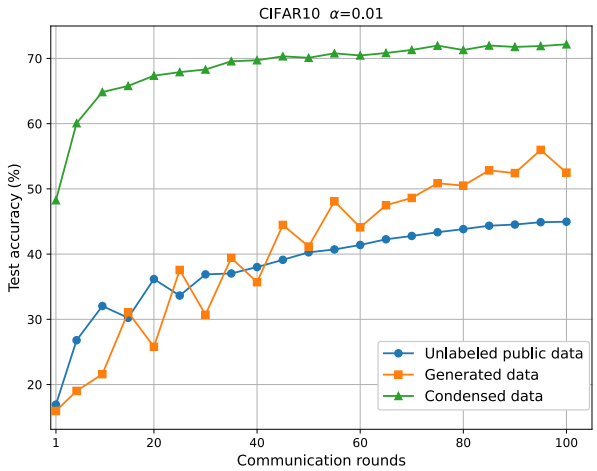


Fig. 3. Comparison of test accuracy of the global model from different forms of data.

Definition 1 (DP [50]): A randomized mechanism M: $D \rightarrow R$ with domain D and range R satisfies (ϵ, δ)-DP if for any two adjacent datasets D_1 and D_2 and any measurable subset $S \subseteq R$

$$P_r(M(D_1) \in S) \leq e^{\epsilon} P_r(M(D_2) \in S) + \delta. \tag{6}$$

Definition 2 (Sensitivity [50]): Let D denote the domain of possible input data and R denote the domain of all possible outputs. The sensitivity of a function f: $D \rightarrow R$ is the maximum amount by which the function value changes when a single entry of the input is perturbed, i.e.,

$$\Delta_p = \max_{D_1, D_2 \subset D} \|f(D_1) - f(D_2)\|_p. \tag{7}$$

TABLE I
SYMBOLS AND THEIR MEANINGS

Symbols	Description
D	Training set
S	Set of synthesized samples
C	Set of classes
K	Number of clients
x	Original data
$\tilde{x}$	Condensed data
$\mathcal{N}$	Standard Gaussian noise
ω	Model parameter
T	Iteration count for distribution matching
P_ω	Probability distributions over parameters
τ_1/τ_2	Distillation temperature
R	Number of communication round
η_c/η_s	Learning rate
y	Ground truth
N	Number of training samples
μ	Mean of feature representations
h_ω	Feature extractor
f_w	Model parameter without last softmax layer
φ	Softmax function
γ	Weight of distillation loss
M	Iteration count for *learn with past*

Definition 3 (Gaussian Mechanism [50]): Let $f : X \rightarrow R_d$ be an arbitrary d-dimensional function with sensitivity defined as

$$\Delta_2 = \max_{D_1, D_2 \subset D} \|f(D_1) - f(D_2)\|_2. \tag{8}$$

For two adjacent datasets D_1 and D_2, the Gaussian mechanism M_σ, parameterized by σ, adds noise into the output,

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

i.e.,

$$M_\sigma(x) = f(x) + \mathcal{N}\left(0, \delta^2 I\right). \tag{9}$$

For numerical data, a Gaussian mechanism defined in [50] can be used to guarantee (ϵ, δ)-DP. We present the DP mechanism by adding artificial Gaussian noise.

Theorem 1 (DP of DDM): There exist two adjacent datasets D_1 and D_2 such that M_σ satisfies (ϵ, δ)-DP for $\geq \sqrt{2ln(1.25/\delta)}\Delta_2/\epsilon$.

Proof: The privacy loss can be expressed as

$$\begin{aligned}
\ln \frac{\Pr(M_\sigma(D_1) = o)}{\Pr(M_\sigma(D_2) = o)} &= \ln \frac{\Pr(f(x_1) + \mathcal{N}(0, \delta^2 I) = o)}{\Pr(f(x_2) + \mathcal{N}(0, \delta^2 I) = o)} \\
&= \ln \frac{\Pr(\mathcal{N}(0, \delta^2 I) = o - f(x_1))}{\Pr(\mathcal{N}(0, \delta^2 I) = o - f(x_2))} \\
&= \ln \frac{\Pr(\mathcal{N}(0, \delta^2 I) = a_1)}{\Pr(\mathcal{N}(0, \delta^2 I) = a_1 + S\ )} \\
&= \ln \left| \frac{\exp((-\frac{1}{2\delta^2})\|a_1\|_2^2)}{\exp((-\frac{1}{2\delta^2})\|a_1 + S\|_2^2)} \right| \\
&= \left| (-\frac{1}{2\delta_t^2})(\|a_1\|_2^2 - \|a_1 + S\|_2^2) \right|
\end{aligned} \tag{10}$$

where $o \in O$ is an output and a_i represents the difference between o and $f(x)$.

According to [50, Th. A.1.], when $\delta = c \times S/\epsilon$, and the constant $c \geq \sqrt{2\ln(1.25/\delta)}$, the protection processing M_σ is (ϵ, δ)-DP. ∎

Assume that the synthetic set can be regarded as an equivalence to the network parameter in DP-SGD [51]. Since the synthetic dataset is initialized from random noise, data privacy will not be leaked at the beginning of the optimization, [51, Th. 1] holds during optimizing S_k. and thus, DDM can ensure that the synthesized data satisfies (ε,δ)-DP by leveraging DP-SGD to update the synthetic dataset, and thus the condensed data does not reveal sensitive information about training samples in the original dataset.

In summary, in a k-client FL system. the design of DDM takes advantage of low DP budgets to achieve comparable performance with a mild utility loss.

2) Collaborated Knowledge Distillation: Because of statistical heterogeneity caused by non-IID data, the cluster centers estimated locally are inevitably biased. Based on the collected condensed dataset $S = \sum_{k=1}^{K}\sum_{c=1}^{C} \widetilde{x}_{k,c}$ uploaded from clients, we propose an efficient and effective KD mechanism, termed CKD. Its key idea is making the ensemble of local models as the teacher's knowledge to guide local model (as a student) learning in the server, which has richer information entropy than the individual local model. Hence, the abundant computing resources of a server may be used to improve the performance of FL. To this end, the procedure of CKD is detailed next.

As the collected condensed dataset S is accessible to all local models, given an instance $\widetilde{x} \in S$, we compute the linear combination of the soft logits provided by local models $\{z_k = f_{[w_k]_0}(\widetilde{x})\}$ and $z = \sum_{k=1}^{K} \alpha_k \cdot z_k$ as a reasonable, high-quality soft target to perform CKD. Here, $\alpha_k = N_k / \sum_{k=1}^{K} N_k$ is a positive number and $\sum_{k=1}^{K} \alpha_k = 1$. N_k is the number of training samples on client k.

Define the collaborative KD loss $\mathcal{L}_{\text{CKD}}^k$

$$\mathcal{L}_{\text{CKD}}^k = \sum_{S} \mathcal{L}_D((\varphi(z, \tau_1)||(\varphi z_k, \tau_1)) \tag{11}$$

where τ_1 is the distillation temperature parameter that controls the shape of the distribution in order to extract richer knowledge from the ensemble output. $\varphi(\cdot)$ is the softmax function used in the final output layer. The Kullback–Leibler divergence is chosen as the KD loss $\mathcal{L}_D$.

By minimizing $\mathcal{L}_{\text{CKD}}^k$ over the condensed dataset S , local model $[w_k]_0$ of client k updates $[w_k]_0$ to $[w_k]_1$. It is worth mentioning that only a few synthesized samples and their labels are shared among these local models. This mechanism has no further supervisory signals and is easy to implement. In addition, the communication cost during CKD process is moderate. Finally, due to the fact that only a few synthesized images are shared, data privacy can be guaranteed.

3) Learn With Past: In the final stage, since condensed data accumulated at the server-side varies in distribution over time, the global model may suffer from severe inter-round forgetting. We propose to encourage the global model to adapt to newly condensed data with the previous one from the last round. At first, the new global model w^r is initialized by weighted aggregating all uploaded local models $\{[w_k]_1\}$ like FedAvg [12]. Let $w^r = \sum_{k=1}^{K} \alpha_k \cdot [w_k]_1$. Then, the server divides shared data S into many batches, assigns each of them to the newly global model and the previous one from the last round, and collects outputs of above-mentioned models on their assigned batches to construct a labeled set of sample pairs, denoted as $S_r = \{(\widetilde{x}, \widetilde{y}_r) \ \ \forall \widetilde{x} \in S\}$, Next, the preliminary global model w^r is iteratively updated as

$$\begin{aligned}
\mathcal{L}_{\text{LwP}} = &\sum_{\widetilde{x} \in S} \gamma \mathcal{L}_D\left(\varphi\left(f_{w^{r-1}}(\widetilde{x}), \tau_2\right)||\varphi(f_{w^r}(\widetilde{x}), \tau_2)\right) \\
&+ (1-\gamma)\mathcal{L}_{\text{CE}}(\varphi(f_{w^r}(\widetilde{x}), \tau_2), y).
\end{aligned} \tag{12}$$

Here, we use the same form of $\mathcal{L}_D$ as in (11). τ_2 is the distillation temperature parameter. γ is a predefined parameter to weight the distillation loss. Note that we only transfer corresponding output of S from the last global model to the new one with only a small amount of communications.

There have been other methods to enhance the performance of the global model, which either updating rely on public data or generated pseudo data. Public data may mislead the knowledge transfer from local models to the aggregated global model, while generated pseudo data cannot cover the real data distribution completely. However, condensed data S provides a more global view of the real data distribution landscape, and the ensemble outputs computed in the CKD stage provides additional opportunities for equal knowledge exchange. On this basis, the aggregated global model can fully learn from local models and the global model in the last round.

We summarize the operations performed by the selected clients and the server at each round. Then, the following processes are repeated for R rounds.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

In each communication round, FedDC first randomly selects a set of clients and distributes the global model to them. Next, local training consists of four stages on the client side: each client 1) initializes the local model parameters with those of the global model ω^{r-1}; 2) performs local DC to obtain condensed data S_k; 3) trains local model $[w_k]_0$ with a local optimizer; and 4) uploads the condensed data S_k and further updates local model $[w_k]_0$ to $[w_k]_1$ by the ensembled logits z from the server. On the server side, the server integrates the local updates (model weights) and aggregates them as a preliminary global model ω^r. Instead of distributing the aggregated global model to each client directly, FedDC enhances this newly formed global model in the server. Concretely, we develop a KD scheme to furtherly explore and transfer the knowledge to the global model. Then, the server can broadcast the updated ω^r to all clients.

In summary, the proposed FedDC framework in each round is realized in Algorithm 1. It is worth noting that except the uploading and receiving of global model parameters, our work only outputs a very small number of synthesis samples with their truth labels between each client and server. In this way it can protect data privacy without significantly increasing the communication cost.

Algorithm 1: FedDC

Input: Input Training set D; number of client K; number of communication round R; set of synthetic samples S; deep neural network parameterized with ω; training iterations of distribution matching T; learning rates η_s and η_c; probability distribution over parameters P_ω; number of local epochs E

Output: Output The final global model ω^R.

Server executes:
for $r = 1, \cdots, R$ **do**
  for client $k = 1, \cdots, K$ **do**
    send the global model ω^r to client k
    $S_k, \omega_k^r \leftarrow$ **ClientUpdate**(k, ω^r)
    transmit S_k, ω_k^r to the server
  end
  aggregate synthesized data $S = S_1 \cup S_2, \ldots S_K$
  for client $k = 1, \cdots, K$ **do**
    compute collaboratively knowledge distillation loss $\mathcal{L}_{\text{CKD}}^k$ based on (11)
    update $\omega_k^r \leftarrow \omega_k^r - \eta_s \nabla_{D_k} \mathcal{L}_{\text{CKD}}^k$
  end
  aggregate local models $\omega_{r+1} \leftarrow \sum_{k=1}^K \frac{|D_k|}{|D|} \omega_k^r$
  compute learn with past loss $\mathcal{L}_{\text{LwP}}$ based on (12)
  update $\omega^{r+1} \leftarrow \omega^{r+1} - \eta_s \nabla_S \mathcal{L}_{\text{LwP}}$
end
ClientUpdate(k, ω^r):
initialize S_k from random noise
for $t = 1, \cdots, T$ **do**
  sample $\omega \sim P_\omega(\omega^r)$
  sample mini-batch pairs $B_c^{D_k} \sim D_k$ and $B_c^{S_k} \sim S_k$ for each class c
  compute differential distribution matching loss $\mathcal{L}_{\text{DDM}}^k$ based on (5)
  update $S_k \leftarrow S_k - \eta_c \nabla_{S_k} \mathcal{L}_{\text{DDM}}^k$
end
return ω^R

C. Time Complexity

As described in Section IV-B, apart from FedAvg [12], the other nine peers are designed for tackling the data heterogeneity issue. MOON [39], FedCurv [52], and FedDM [35] obtain relatively excellent performance (in Section IV-B). We thus compare FedDC with them. The time complexity of FedDC is

$$O(\text{FedDC}) = O(R(K(T+E)+M)). \tag{13}$$

From [35], [39], and [52], we have

$$\begin{aligned} O(\text{MOON}) &= O(R \cdot K \cdot E) \\ O(\text{FedCurv}) &= O(R \cdot K \cdot E) \\ O(\text{FedDM}) &= O(R(K \cdot T + M)). \end{aligned}$$

Hence, we conclude that MOON [39] and FedCurv [52] are the fastest with the same time complexity while FedDC exhibits a relatively higher computational complexity.

IV. Experiments and Results

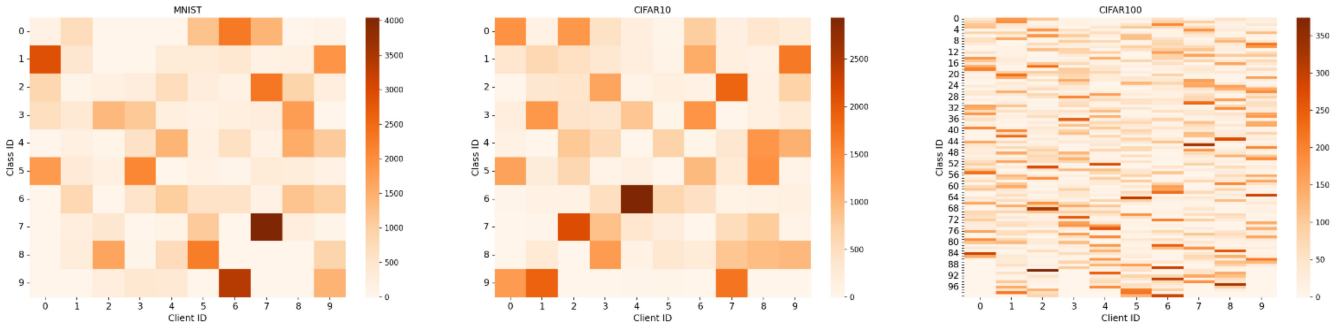
A. Experimental Setup

1) Datasets: In this article, we conduct experiments to compare and evaluate the performance of FedDC on label-skew data heterogeneity. We adopt MNIST [53], CIFAR10 [54], and CIFAR100 [54]. The MNIST dataset consists of 70 000 grayscale images that fall into ten categories. The CIFAR-10 and CIFAR-100 datasets are both collections of 60 000 color images. CIFAR10 images are grouped into ten classes with 6000 images per class (IPC), while CIFAR100 categorizes image into 100 classes with 600 IPC.

2) Comparison Algorithms: We compare FedDC against FedAvg [12] and various SOTA FL algorithms designed for handling data heterogeneity: FedProx [14], FedNTD [40], MOON [39], FedDyn [41], FedNova [55], SCAFFOLD [15], and FedCurv [52]. We also consider recent work FedDM [35] that employs DC to demonstrate the effectiveness of our proposed schemes.

3) Hyperparameters: For both FedDC and FedDM [35], following [18], in each communication round, the batch size of real images is set to 256, we tune the number of IPC in {1, 5, 10, 20, 50, 80, 100}, and use 1000 iterations with learning rate of 1.0 for synthetic set S_k. Suggested by [19], each class of synthetic images is initialized as the average of randomly sampled real images with corresponding real labels. If considering privacy protection, random noise initialization is added to synthetic images. By considering the tradeoff between global model performance and communication efficiency, we choose IPC to be 5 for CIFAR100, 10 for MNIST and CIFAR10 when the number of clients $K = 10$. The choice of radius ρ,

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 4. Clients' data distribution under non-IID data partition $\alpha = 0.01$. Each rectangle denotes the amount of data samples of a specific class located in a client. (a) MMIST. (b) CIFAR10. (c) CIFAR100.

TABLE II
TEST ACCURACY OF FL METHODS WITH DIFFERENT LEVEL OF NONUNIFORM DATA PARTITIONING

Method	MNIST		CIFAR10		CIFAR100	
	β=0.01	β=0.10	β=0.01	β=0.10	β=0.01	β=0.10
FedAvg [12]	92.64±1.87	97.68±0.25	55.55±4.03	71.67±1.37	45.55±1.03	48.66±0.20
FedCurv [52]	93.00±2.29	97.80±0.28	55.41±1.30	71.98±2.13	45.65±0.68	48.47±0.25
FedDyn [41]	57.28±17.95	96.66±0.54	28.44±6.53	67.63±5.63	43.26±0.25	48.00±1.58
FedNTD [40]	86.60±1.54	97..47±0.17	50.01±4.47	70.24±1.17	43.42±0.55	49.77±0.27
FedProx [14]	92.33±0.68	97.61±0.27	53.49±0.68	71.71±1.39	45.36±0.61	48.50±0.26
MOON [39]	93.20±0.98	97.78±0.24	54.55±4.18	71.99±1.36	45.64±0.43	48.97±0.14
FedNova [55]	90.20±1.57	97.77±0.12	52.89±3.59	70.69±2.50	45.89±0.65	48.86±0.50
SCAFFOLD [15]	93.34±2.00	97.67±0.30	54.37±0.72	71.73±1.45	45.44±0.87	45.44±0.87
FedDM [35]	99.14±0.03	99.07±0.01	66.98±0.36	71.62±0.41	46.05±0.53	50.07±0.16
FedDC(Ours)	**99.16±0.04**	**99.25±0.05**	**69.84±0.22**	**74.43±0.40**	**48.69±0.15**	**50.80±0.40**

that is used to clip the sampled weight of a global model to guarantee $\|\omega - \omega_r\|_2 \leq \rho$, is tested and discussed later. On the server side, the global model is trained for 500 epochs by SGD of $\eta_s = 0.01$ with the batch size 256. For other comparison algorithms, we also choose the same batch size of 256 for local updating and adopt 5 local epochs with the same learning rate η_s. For a fair comparison, all FL methods are finished with the same number of communication rounds 100 by default, and the general model structure ConvNet [18] for all experiments. Three random seeds are required for the experiments with two NVIDIA 4090Ti GPUs. The mean testing accuracy of global model is reported. We partition the training split of each benchmark dataset into multiple data shards to simulate the local training dataset for every client. Specifically, we adopt the latent Dirichlet allocation (LDA) [42] non-IID partition strategy, which assigns partition of class c by sampling $p_c \approx Dir(\alpha)$. Through it both distribution and dataset size are different for each client. For every benchmark dataset, we consider three degrees of data heterogeneity, represented by $\alpha = 0.01$ and $\alpha = 0.1$, respectively. The hyperparameter α controls the strength of heterogeneity. Notably, a smaller α implies a higher non-IID in the data distribution among clients. We choose these α values to simulate harsh scenarios of data heterogeneity that can be encountered in real world applications. In this way, each client can onlyhave relatively few unbalanced data samples in certain classes with the above data partitioning strategy. The data distributions among parties in default settings are shown in Fig. 4.

B. Accuracy Comparison

Table II shows the top-1 test accuracy of the global model achieved by all approaches in various settings. Comparing different FL approaches, Table II shows that FedDC is consistently the best approach in both mean accuracy and variance among all tasks. A detailed analysis in Table II indicates that FedDC can outperform FedAvg by up to 6.52%, 14.29%, and 3.14% on MNIST, CIFAR10, and CIFAR100, and respectively. FedDM also employs DM to enhance the global model on the server side. It outperforms FedNova, FedNTD, FedCurv, FedProx, and FedDyn, and relatively surpasses MOON and SCAFFOLD in some cases, but it is worse than our proposed method FedDC. The performance of further verifies the effectiveness of the proposed DDM module on the global view of data distribution landscape. Substantially, FedDC's performance advantage is more obviously shown under stronger data heterogeneity, such as at α=0.01, demonstrating the effectiveness of our CKD and LwP fundamental techniques.

C. Convergence Performance

We evaluate and compare our method with other FL algorithms on all three datasets in various settings. The learning curves are, respectively, illustrated in Fig. 5. We can conclude that FedDC can get faster convergence and significantly outperforms its peers by a large margin. Particularly, we examine and compare FedDC with others in convergence speed under extreme data heterogeneity. For instance, at $\alpha = 0.01$, FedDC

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EVALUATION OF DIFFERENT FL METHODS IN TERMS OF COMMUNICATION COST (MB) TO REACH TARGET TEST ACCURACY (acc)

Method	MNIST		CIFAR10		CIFAR100	
	acc=57% / β=0.01	acc=96% / β=0.10	acc=50% / β=0.01	acc=67% / β=0.10	acc=43% / β=0.01	acc=45% / β=0.10
FedAvg [12]	25.98±6.58	56.75±11.30	25.29±12.53	90.23±16.41	123.72±14.40	99.26±0.37
FedCurv [52]	28.71±2.05	53.32±14.79	23.93±3.14	87.51±20.65	121.67±10.32	99.26±0.21
FedDyn [41]	28.71±12.47	117.57±37.22	24.61±8.20	91.26±7.26	145.61±12.47	97.06±0.76
FedNTD [40]	21.18±2.36	61.52±3.55	19.83±1.19	100.49±17.76	176.37±0.00	101.62±0.41
FedProx [14]	24.61±4.10	55.37±10.85	21.88±12.53	84.76±17.68	136.03±19.05	99.09±0.43
MOON [39]	27.34±5.17	54.00±10.52	20.51±12.47	93.66±17.92	123.72±17.08	99.93±0.39
FedNova [55]	30.76±12.47	67.68±19.77	29.39±5.17	118.95±6.15	117.57±13.66	100.22±0.88
SCAFFOLD [15]	25.29±1.19	56.05±12.02	27.34±11.83	90.23±16.41	123.72±18.60	93.13±1.58
FedDM [35]	2.65±0.00	2.65±0.00	4.39±0.00	60.64±5.83	73.01±7.01	17.57±0.00
FedDC(Ours)	**1.62±0.00**	**1.62±0.00**	**3.37±0.00**	**23.42±2.55**	**38.10±2.55**	**16.85±0.00**

achieves the highest accuracy in initial few rounds. Compared to FedDM on CIFAR10 dataset, FedDC achieves round 65% on test accuracy after 20 communication rounds, while the second best, i.e., FedDM, has nearly 60% only under the same conditions.

D. Communication Cost

To illustrate FedDC has higher communication efficiency than other FL approaches, we compare and analyze the communication cost of our proposed method. Take the three datasets as examples, we conduct experiments by evaluating communication cost to reach the target test accuracy in Table III. For FedDC, we need to download the global model, and upload the condensed data and updated model parameters. According to Table III, we notice that FedDC requires much lower communication cost than its peers, since the design of CKD and LwP modules can guarantee to effectively transfer knowledge from local models to the preliminary global model. This also shows that FedDC is more suitable for low bandwidth scenarios.

E. Statistical Analysis

Here, we compare FedDC with other approaches in terms of final accuracy on CIFAR10 dataset under the data partitioning of $\alpha = 0.01$ by using t-test and Wilcoxon's test. $P_{\text{Wilconxon}}$ and $P_{t-\text{test}}$ represent the p-value of t-test and Wilcoxon's test, respectively. Based on the analysis of results in Table IV. When $P_{t-\text{test}} < 0.05$ and $P_{\text{Wilconxon}} > 0.05$, it denotes that FedDC performs significantly better than others in terms of final accuracy.

TABLE IV
RESULTS OF t-TEST AND WILCOXON'S TEST OF FEDDC VERSUS ITS PEERS IN ACCURACY ("+" = FEDDC IS SIGNIFICANTLY BETTER)

Configuration	$P_{\text{Wilconxon}}$	$P_{t-\text{test}}$	Result
FedDC vs FedAvg [12]	0.0000	0.0268	+
FedDC vs FedCurv [52]	0.0000	0.0030	+
FedDC vs FedDyn [41]	0.0000	0.0078	+
FedDC vs FedNTD [40]	0.0000	0.0158	+
FedDC vs FedProx [14]	0.0000	0.0012	+
FedDC vs MOON [39]	0.0000	0.0242	+
FedDC vs FedNova [55]	0.0000	0.0127	+
FedDC vs SCAFFOLD [15]	0.0000	0.0015	+
FedDC vs FedDM [35]	0.0000	0.0078	+

F. Parameter Sensitivity

In this section, we have a detailed analysis of FedDC to demonstrate effects of factors, including the number of local epochs E, IPC, noise level σ, network structure, and selection of ρ-radius ball.

1) Number of Local Epochs: As seen in Fig. 6, given the same number of communication rounds, when the number of local epochs is set to 1, the accuracy of the global model is relatively low and the learning curve fluctuations up and down. Even so, FedDC is still the best in both stability and accuracy. As the number of local epochs scale up to 5, the final accuracy of all approaches rises. Similarly, FedDC obviously outperforms its peers. All this indicates that FedDC can relieve the negative effects of the model drift caused by too many local updates on the server side.

2) Visualization of Synthetic Dataset: We conduct experiments on CIFAR10 dataset under the data partitioning of $\alpha = 0.5$ and condensed data S is initialized from random Gaussian noise $\mathcal{N}(0,\ 1)$. Fig. 7(a)–(c) shows the synthetic dataset under different noise levels. Even when there is no noise added to the gradient during optimization of the synthetic dataset. It can be observed that those synthetic images are harder to recognize their original classes. Of course, as σ grows up, synthesized data becomes more illegible. This further verifies that client's data privacy is strongly protected. Furthermore, we find from Fig. 7(d) that the learning curves of FedDC with different degrees of noise are of similar shape, and the top-1 final test accuracy of global models changes only slightly.

3) Effects of IPC: With different IPC values, we provide the result of top-1 final accuracy of FedDC on CIFAR10 dataset under various degrees of data heterogeneity in Table V. When the IPC values scaling up, the final accuracy of the global model grows. Considering client's privacy and communication cost with the value of IPC, an IPC value lower than

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

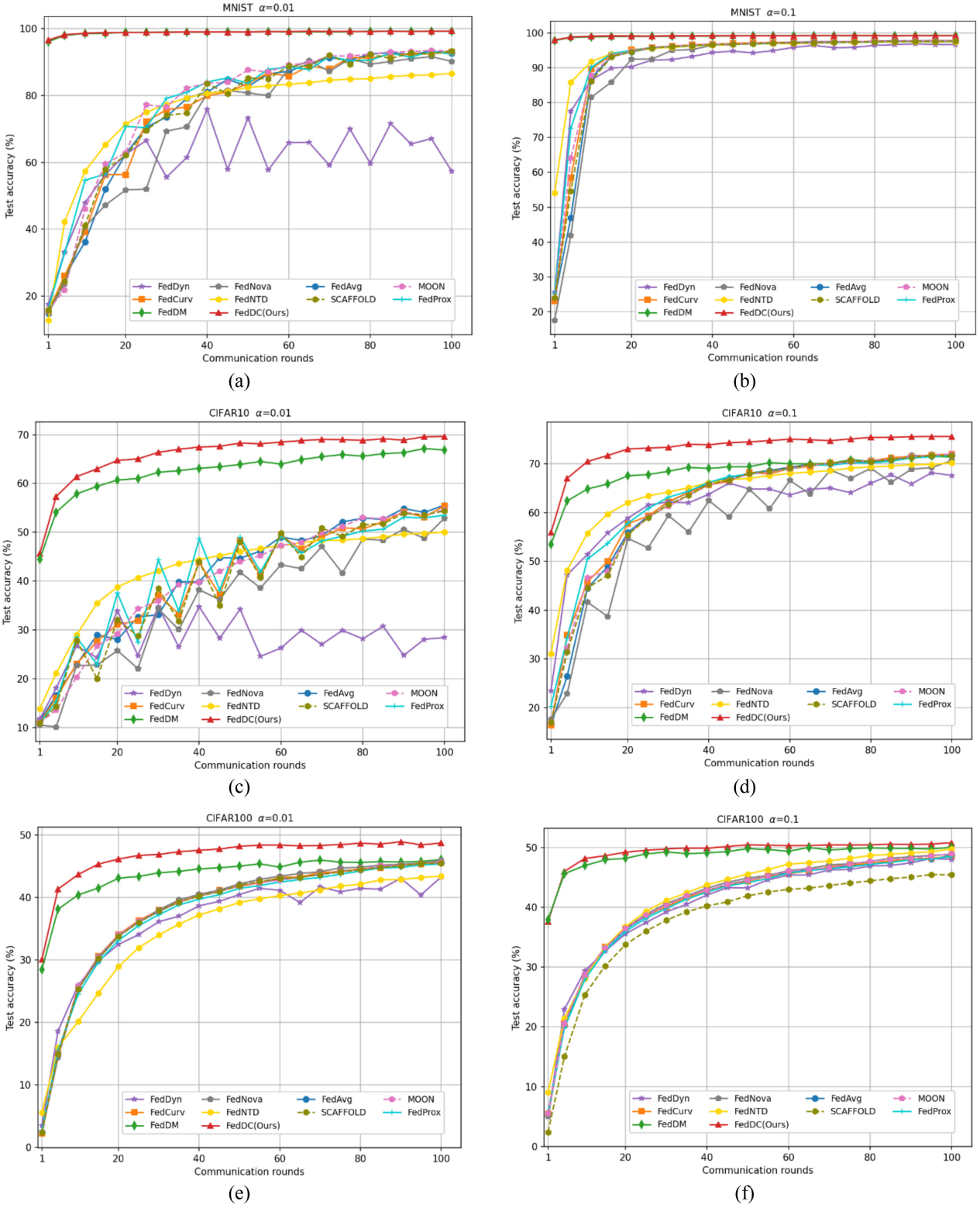


Fig. 5. Comparison of Top-1 test accuracy along with the number of communication rounds on (a) and (b) MNIST, (c) and (d) CIFAR10, and (e) and (f) CIFAR100. In terms of convergence speed and final test accuracy, FedDC outperforms its peers in all three datasets, especially on the datasets with high data heterogeneity.

50 should be considered as a tradeoff among performance, communication cost, and privacy. Specially, we also consider an IID situation. Our result further validates the compatibility and scalability of FedDC in different FL scenarios.

4) Different Network Structures: Besides ConvNet, for a more complicated and larger model ResNet-18, we evaluate FedDC on CIFAR10 dataset under the default setting $\alpha = 0.01$. As seen in Fig. 8, FedDC surpasses its peers. This further

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

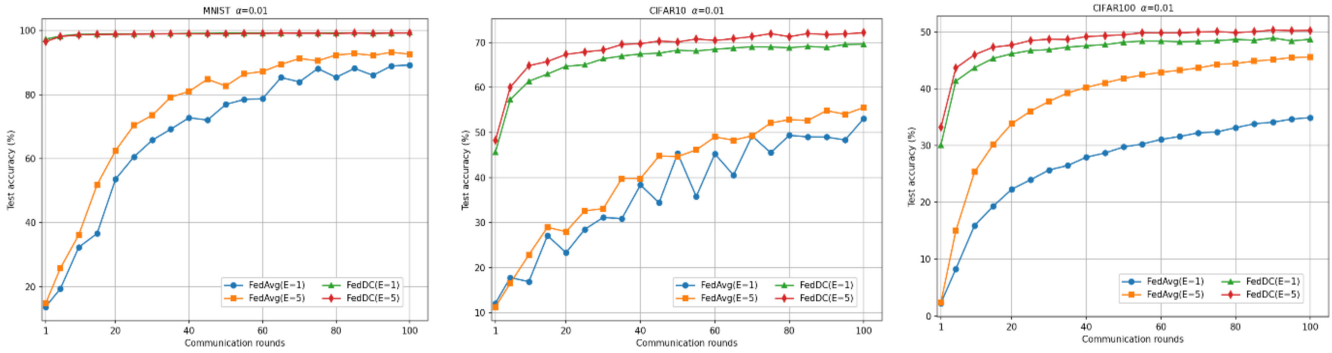


Fig. 6. Top-1 test accuracy with different number of local epochs on three datasets.

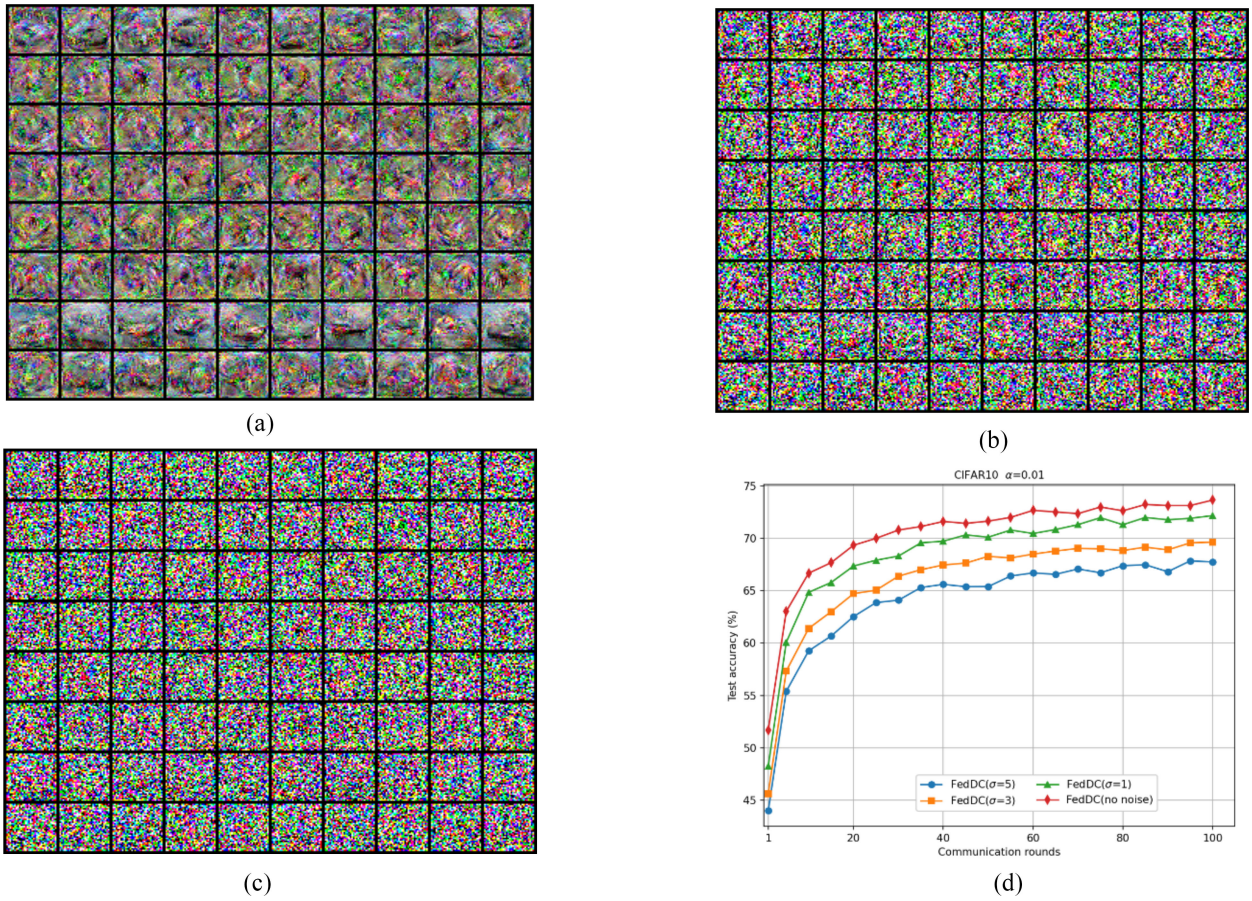


Fig. 7. Visualization of the synthetic dataset and comparison of convergence performance amongst comparison approaches. (a)–(c) Synthetic dataset with different level of noise. (d) Performance of FL methods without or with noise.

verifies that FedDC is more efficient in larger networks, since it is CKD and LwP modules guarantee efficient communication during training.

5) Selection of ρ-Radius Ball: To investigate the effect of different value ρ, we conduct the experiments by choosing ρ from [2, 3, 5, 10] and presenting the final accuracy of the global model under the data partition of $\alpha = 0.01$ and 0.1 after 100 communication rounds. According to Table VI, we notice that test accuracies are similar on CIFAR10 dataset, Thus, it can be concluded that FedDC is not very sensitive to the choice of ρ. The same conclusion holds true for an IID FL scenario, and we decide to select $\rho = 5$ for all our experiments.

G. Ablation Analysis

We conduct experiments to investigate how they affect the improvement of final accuracy on CIFAR10 dataset. Here, we compare our proposed method FedDC with two other aforementioned configurations where each fundamental design is not used. We also compare and analyze with FedDM. As shown in Table VII, "w/o CKD" denotes the FedDC without collaborative KD, "w/o LwP" denotes the

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

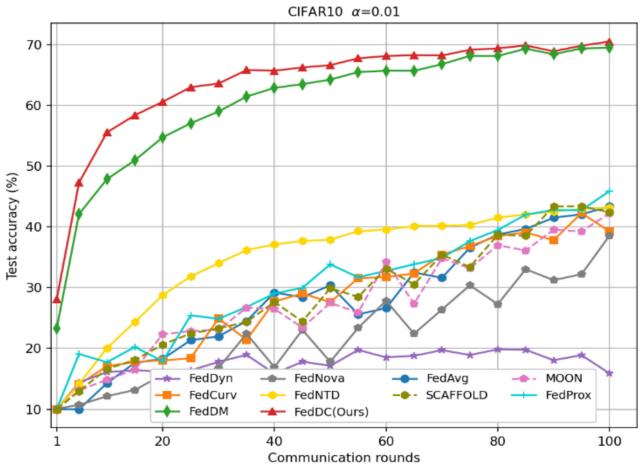


Fig. 8. Top-1 test accuracy on ResNet-18.

TABLE V
IMPACT OF IPC ON THE GLOBAL MODEL FINAL ACCURACY

Configuration	IID	β=0.01	β=0.10
IPC=1	79.65±0.34	49.34±0.54	72.30±0.40
IPC=5	79.06±0.16	59.76±1.22	71.29±0.61
IPC=10	79.18±0.02	63.75±0.37	72.46±0.68
IPC=20	80.34±0.25	67.31±0.35	74.16±0.51
IPC=50	80.84±0.65	69.20±0.37	69.38±1.50
IPC=80	81.14±0.31	70.05±0.28	75.67±0.11
IPC=100	81.42±0.22	70.45±0.19	74.60±1.25

TABLE VI
RESULTS OF TEST ACCURACY OF FEDDC UNDER DIFFERENT ρ

ρ	IID	β=0.01	β=0.10
$\rho = 2$	79.60±0.53	63.40±0.11	72.46±0.40
$\rho = 3$	79.53±0.21	63.40±0.31	72.33±0.66
$\rho = 5$	79.18±0.02	63.75±0.37	72.46±0.68
$\rho = 10$	79.76±0.23	64.24±0.18	72.73±0.45

FedDC without LwP. The average testing accuracy obtained from only one of the basic techniques still shows a significant improvement over FedDM [35]. We add an ablation experiment on the DDM component in Fig. 9. The global model suffers from performance degradation when clients directly uploaded raw images with added noise. Moreover, the complete FedDC shows the greatest improvement in average testing accuracy. The same conclusion holds true for an IID FL scenario. To confirm our successful solution to the issue of forgetting, we measure the similarity of class-wise test accuracy to the previous round as the communication rounds proceed. From Fig. 10, we can see that FedDC ensures high consistency of class-wise test accuracy between two adjacent rounds. This clearly shows that the catastrophic forgetting problem has been successfully solved. To conclude, FedDC indeed can mitigate the catastrophic forgetting and data heterogeneity by promoting the utilization of additional knowledge derived from local models and global model, and thus, enhances the overall learning performance.

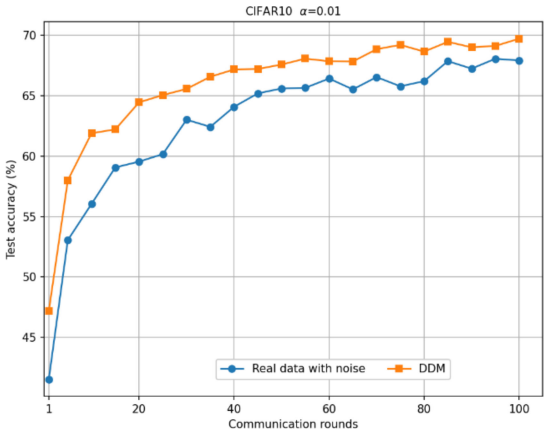


Fig. 9. Performance of FedDC with DDM and real data with noise.

TABLE VII
IMPACT OF CORE DESIGN ON THE GLOBAL MODEL ACCURACY FOR LEARNING CIFAR10 ON CONVNET

Configuration	IID	β=0.01	β=0.10
FedDC	79.18±0.02	63.75±0.37	72.46±0.68
w/o CKD	77.42±0.46	61.76±0.69	70.60±0.19
w/o LwP	79.14±0.42	63.42±0.38	72.36±0.40
FedDM [35]	68.67±0.31	57.92±0.73	63.53±0.34

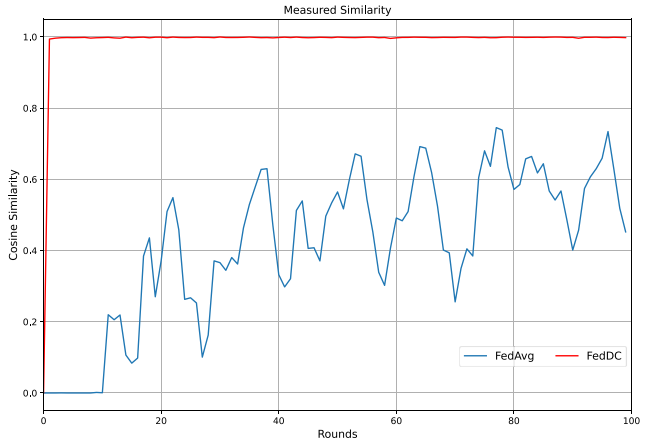


Fig. 10. Comparison of cosine similarity among FL methods.

H. IoT Dataset

The Army Cyber Institute (ACI) IoT Network Traffic Dataset 2023 (ACI-IoT-2023) is a comprehensive dataset focusing on IoT network security [56], It is collected from a highly simulated home IoT environment which covers diverse and realistic network traffic data over five days, including reconnaissance, denial of service, brute force, spoofing attacks, and normal network traffic. To achieve high anomaly detection accuracy, we adopt a simplified residual neural network model (LwResnet) [57], which is based on the improvement of deep 1-D CNN for traffic characteristic data. Apart from FedAvg [12], we also compare a baseline approach named *Local* where each party trains a model with its local data without model aggregation. We examine and compare FedDC with them in terms of final anomaly detection accuracy. As

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

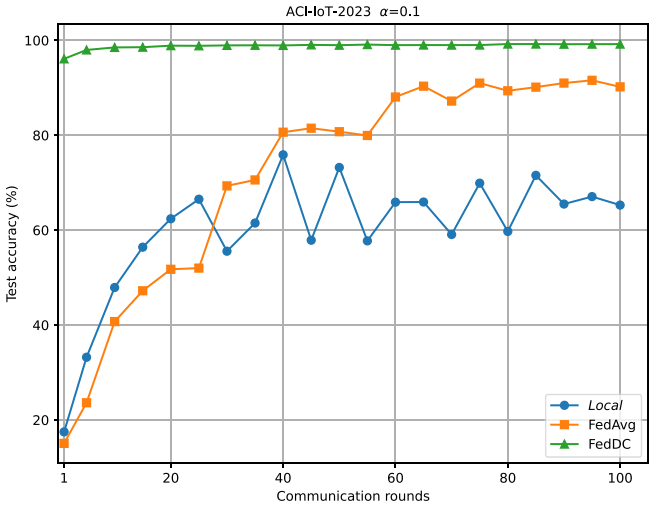


Fig. 11. Test accuracy *Local*, FedAvg, and FedDC on the ACI-IoT-2023 dataset.

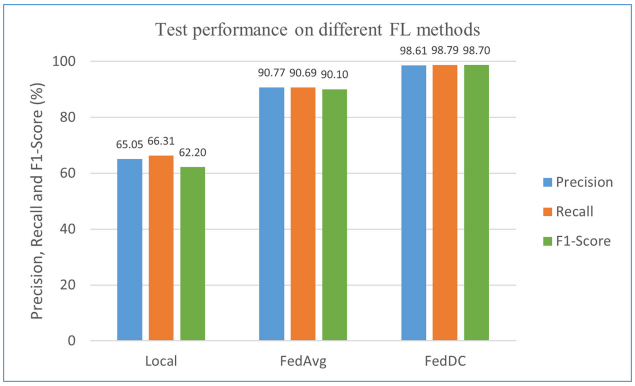


Fig. 12. Final performance of different FL methods on the ACI-IoT-2023 dataset.

shown in Fig. 11, *Local* demonstrates much worse accuracy than other FL approaches. This demonstrates the benefits of FL. Compared to FedAvg [12], FedDC shows a big improvement, which is mainly due to our alignment of new knowledge and mitigation of previous ones in FedDC. Additional experimental comparison results in terms of Precision, Recall and F1-Score are shown in Fig. 12. From it, FedDC consistently stands out as the best approach.

V. Conclusion

In this article, we propose FedDC as a novel FL approach for vision tasks in realistic IoT scenarios, which intends to mitigate both data heterogeneity and catastrophic forgetting issues among intradomains. By learning from local models and the global model based on the collected condensed data, 1) clients can learn from its peers and 2) the server is bound to improve the performance of the aggregated global model. Consequently, FedDC demonstrates considerable improvement over SOTA FL approaches in terms of convergence speed and classification accuracy. As for future work, we intend to improve high-dimensional heterogeneous knowledge transfer [58] across clients by structural KD [59], [60], [61], [62], [63], [64], which represents an important method in knowledge transfer.

VI. Acknowledgment

We are also grateful for the efforts from our colleagues in the Sino-German Center of Intelligent Systems, Tongji University.

References

[1] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[2] H. Zhang, T. Wu, S. Cheng, and J. Liu, "FedCos: A scene-adaptive enhancement for federated learning," *IEEE Internet Things J.*, vol. 10, no. 5, pp. 4545–4556, Mar. 2023.
[3] H. Mu, G. Zhang, Z. Ma, M. Zhou, and Z. Cao, "Dynamic obstacle avoidance system based on rapid instance segmentation network," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 5, pp. 4578–4592, May 2024.
[4] Q. Xie et al., "Efficiency optimization techniques in privacy-preserving federated learning with homomorphic encryption: A brief survey," *IEEE Internet Things J.*, vol. 11, no. 14, pp. 24569–24580, Jul. 2024.
[5] W. Huang et al., "Federated learning for generalization, robustness, fairness: A survey and benchmark," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9387–9406, Dec. 2024.
[6] L. Yuan, Z. Wang, L. Sun, P. S. Yu, and C. G. Brinton, "Decentralized federated learning: A survey and perspective," *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34617–34638, Nov. 2024.
[7] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, May 2020.
[8] A. Rauniyar et al., "Federated learning for medical applications: A taxonomy, current trends, challenges, and future research directions," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7374–7398, Mar. 2024.
[9] W. Ni, H. Ao, H. Tian, Y. C. Eldar, and D. Niyato, "FedSL: Federated split learning for collaborative healthcare analytics on resource-constrained wearable IoMT devices," *IEEE Internet Things J.*, vol. 11, no. 10, pp. 18934–18935, May 2024.
[10] T. Kim, E. Lin, J. Lee, C. Lau, and V. Mugunthan, "Navigating data heterogeneity in federated learning: A semi-supervised approach for object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–8.
[11] L. Zhong and K. Liu, "Visual classification and detection of power inspection images based on federated learning," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 5460–5469, Jul./Aug. 2024.
[12] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Stat.*, 2017, pp. 1273–1282.
[13] P. Kairouz et al., "Advances and open problems in federated learning," *Found. Trends Mach. Learn.*, vol. 14, no. 1, pp. 1–210, 2021.
[14] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 2, 2020, pp. 429–450.
[15] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "Scaffold: Stochastic controlled averaging for federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 5132–5143.
[16] H. Zhang, S. Li, P. Wang, D. Zeng, and S. Ge, "M3D: Dataset condensation by minimizing maximum mean discrepancy," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 9314–9322.
[17] B. Zhao and H. Bilen, "Dataset condensation with differentiable Siamese augmentation," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12674–12685.
[18] B. Zhao and H. Bilen, "Dataset condensation with distribution matching," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2023, pp. 6503–6512.
[19] B. Zhao, K. R. Mopuri, and H. Bilen, "Dataset condensation with gradient matching," in *Proc. Int. Conf. Learn. Rep.*, 2021, p. 12.
[20] Y. He, L. Xiao, J. T. Zhou, and I. Tsang, "Multisize dataset condensation," in *Proc. Int. Conf. Learn. Represent.*, 2024, p. 12.
[21] S. Shao, Z. Zhou, H. Chen, and Z. Shen, "Elucidating the design space of dataset condensation," 2024, *arXiv:2404.13733*.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.

[22] G. Cazenavette, T. Wang, A. Torralba, A. A. Efros, and J.-Y. Zhu, "Dataset distillation by matching training trajectories," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4750–4759.
[23] Z. Lu, J. Wang, and C. Jiang, "Data-free knowledge filtering and distillation in federated learning," *IEEE Trans. Big Data*, vol. 11, no. 3, pp. 1128–1143, Jun. 2025.
[24] C. Sun, T. Jiang, and D. Pompili, "Heterogeneous federated learning via generative model-aided knowledge distillation in the edge," *IEEE Internet Things J.*, vol. 12, no. 5, pp. 5589–5599, Mar. 2025.
[25] S. Yao, Q. Kang, M. Zhou, M. J. Rawa, and A. Albeshri, "Discriminative manifold distribution alignment for domain adaptation," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 2, pp. 1183–1197, Feb. 2023.
[26] J. Tang, Q. Kang, M. Zhou, H. Yin, and S. Yao, "MemeNet: Toward a reliable local projection for image recognition via semantic featurization," *IEEE Trans. Image Process.*, vol. 33, pp. 1670–1682, 2024.
[27] A. Mora, I. Tenison, P. Bellavista, and I. Rish, "Knowledge distillation for federated learning: A practical guide," 2022, *arXiv:2211.04742*.
[28] L. Zhang, L. Shen, L. Ding, D. Tao, and L.-Y. Duan, "Fine-tuning global model via data-free knowledge distillation for non-IID federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10174–10183.
[29] J. Liu, Q. Zeng, H. Xu, Y. Xu, Z. Wang, and H. Huang, "Adaptive block-wise regularization and knowledge distillation for enhancing federated learning," *IEEE/ACM Trans. Netw.*, vol. 32, no. 1, pp. 791–805, Feb. 2024.
[30] Y. Wang, H. Fu, R. Kanagavelu, Q. Wei, Y. Liu, and R. S. M. Goh, "An aggregation-free federated learning for tackling data heterogeneity," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26233–26242.
[31] H. Wang, Y. Li, W. Xu, R. Li, Y. Zhan, and Z. Zeng, "DaFKD: Domain-aware federated knowledge distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 20412–20421.
[32] Z. Zhu, J. Hong, and J. Zhou, "Data-free knowledge distillation for heterogeneous federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12878–12889.
[33] J. Zhang et al., "DENSE: Data-free one-shot federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 21414–21428.
[34] S. Wang, Y. Fu, X. Li, Y. Lan, and M. Gao, "DFRD: Data-free robustness distillation for heterogeneous federated learning," in *Proc. 37th Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–13.
[35] Y. Xiong, R. Wang, M. Cheng, F. Yu, and C.-J. Hsieh, "FedDM: Iterative distribution matching for communication-efficient federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 16323–16332.
[36] Y. Ma, Z. Xie, J. Wang, K. Chen, and L. Shou, "Continual federated learning based on knowledge distillation," in *Proc. 31st Int. Conf. Artif. Intell. (IJCAI)*, 2022, pp. 2182–2188.
[37] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 2351–2363.
[38] H. Wang, M. Yurochkin, Y. Sun, D. Papailiopoulos, and Y. Khazaeni, "Federated learning with matched averaging," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–16.
[39] Q. Li, B. He, and D. Song, "Model-contrastive federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10708–10717.
[40] G. Lee, M. Jeong, Y. Shin, S. Bae, and S.-Y. Yun, "Preservation of the global knowledge by not-true distillation in federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 38461–38474.
[41] C. Jin, X. Chen, Y. Gu, and Q. Li, "FedDyn: A dynamic and efficient federated distillation approach on recommender system," in *Proc. IEEE 28th Int. Conf. Parallel Distrib. Syst. (ICPADS)*, 2023, pp. 786–793.
[42] T.-M. H. Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," 2019, *arXiv:1909.06335*.
[43] T. Wang, J.-Y. Zhu, A. Torralba, and A. A. Efros, "Dataset distillation," 2018, *arXiv:1811.10959*.
[44] J. Cui, R. Wang, S. Si, and C.-J. Hsieh, "DC-BENCH: Dataset condensation benchmark," in *Proc. Adv. Neural Inf. Process. Syst*, vol. 35, Dec. 2022, pp. 810–822.
[45] D. Zhang, J. Le, N. Mu, G. Liu, and T. Xiang, "Privacy-preserving federated learning based on dataset condensation," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 748–760, Feb. 2025.
[46] X. Wang, S. Sha, and Y. Sun, "DCFL: Non-IID awareness dataset condensation aided federated learning," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–8.
[47] P. Liu, X. Yu, and J. T. Zhou, "Meta knowledge condensation for federated learning," Sep. 29, 2022, *arXiv:2209.14851*.
[48] G. Hinton, "Distilling the knowledge in a neural network," in *Proc. IEEE NIPS Workshops*, Montreal, QC, Canada, Dec. 2015, pp. 1–9.
[49] W. Zhang, X. Liu, and S. Tarkoma, "FedGK: Communication-efficient federated learning through group-guided knowledge distillation," *ACM Trans. Internet Technol.*, vol. 24, no. 4, pp. 1–21, 2024.
[50] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Found. Trends Theor. Comput. Sci.*, vol. 9, nos. 3–4, pp. 211–407, 2014.
[51] M. Abadi et al., "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conf. Comput. Commun. Security*, 2016, pp. 308–318.
[52] N. Shoham et al., "Overcoming forgetting in federated learning on non-iid data," in *Proc. NeurIPS Workshop Feder. Learn. Data Privacy Confidentiality*, 2019, p. 9.
[53] L. Deng, "The MNIST database of handwritten digit images for machine learning research [best of the Web]," *IEEE Signal Process. Mag.*, vol. 29, no. 6, pp. 141–142, Nov. 2012.
[54] A. Krizhevsky and G. Hinton. "Learning multiple layers of features from tiny images." 2009. [Online]. Avaialble: https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf
[55] J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, "Tackling the objective inconsistency problem in heterogeneous federated optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 7611–7623.
[56] E. A. Nack, M. C. McKenzie, and N. D. Bastian, "ACI-IoT-2023: A robust dataset for Internet of Things network security analysis," in *Proc. IEEE Mil. Commun. Conf. (MILCOM)*, Oct. 2024, pp. 1–6.
[57] Q. Tian, C. Guang, C. Wenchao, and W. Si, "A lightweight residual networks framework for DDoS attack classification based on federated learning," in *Proc. IEEE INFOCOM Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, 2021, pp. 1–6.
[58] M. Jiang et al., "FedHiDiC: High-dimensional heterogeneous data condensation and distillation contrastive learning in federated learning," in *Proc. IEEE Int. Conf. Syst. Man Cybern. (SMC)*, Oct. 2024, pp. 39–44.
[59] S. Yang et al., "Learning from human educational wisdom: A student-centered knowledge distillation method," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 6, pp. 4188–4205, Jun. 2024.
[60] S. Yang et al., "Skill-transferring knowledge distillation method," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6487–6502, Nov. 2023.
[61] K. Liu, Z. Huang, C.-D. Wang, B. Gao, and Y. Chen, "Fine-grained learning behavior-oriented knowledge distillation for graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9422–9436, May 2025.
[62] Z. Huang et al., "Feature map distillation of thin nets for low-resolution object recognition," *IEEE Trans. Image Process.*, vol. 31, pp. 1364–1379, 2022.
[63] N. Zeng, X. Li, P. Wu, H. Li, and X. Luo, "A novel tensor decomposition-based efficient detector for low-altitude aerial objects with knowledge distillation scheme," *IEEE/CAA J. Automatica Sinica*, vol. 11, no. 2, pp. 487–501, Feb. 2024.
[64] L. Xu, J. Ren, Z. Huang, W. Zheng, and Y. Chen, "Improving knowledge distillation via head and tail categories," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3465–3480, May 2024.



Jun Ma (Member, IEEE) received the B.S. degree in computer science and technology from Dalian University of Technology, Dalian, China, in 2018, and the M.S. degree in software engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2021. He is currently pursuing the Ph.D. degree in control science and engineering with Tongji University, Shanghai, China.

His current research interests include federated learning and continual learning.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.



Qi Kang (Senior Member, IEEE) received the B.S. degree in automatic control and the M.S. and Ph.D. degrees in control theory and control engineering from Tongji University, Shanghai, China, in 2002, 2005, and 2009, respectively.

He is currently a Professor with the Department of Control Science and Engineering, Tongji University, and also the State Key Laboratory of Autonomous Intelligent Unmanned Systems, Shanghai. His interests are in swarm intelligence, evolutionary computation, machine learning, and intelligent control and optimization.

Prof. Kang served as the General Chair for the 19th IEEE International Conference on Networking, Sensing and Control. He is currently an Associate Editor of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS and IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS. He is the Secretary General of Shanghai Association for System Simulation.



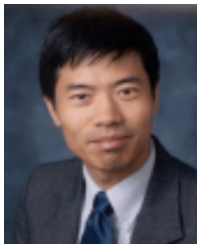
Kefan Wang (Student Member, IEEE) received the B.S. degree in electric power engineering from Shanghai University of Electric Power, Shanghai, China, in 2019, and the M.S. degree in control science and engineering from Shanghai Institute of Technology, Shanghai, in 2022. He is currently pursuing the Ph.D. degree with Shanghai Research Institute for Intelligent Autonomous Systems, Tongji University, Shanghai.

His research interests include machine learning, generative artificial intelligence, and spatiotemporal prediction.



Ruijing Sun (Student Member, IEEE) received the B.S. degree in automation from the Ocean University of China, Qingdao, China, in 2019. She is currently pursuing the M.S. degree in control theory and control engineering with Tongji University, Shanghai, China.

Her research interests include deep learning and image generation.



MengChu Zhou (Fellow, IEEE) received the B.S. degree in control engineering from Nanjing University of Science and Technology, Nanjing, China, in 1983, the M.S. degree in automatic control from Beijing Institute of Technology, Beijing, China, in 1986, and the Ph.D. degree in computer and systems engineering from Rensselaer Polytechnic Institute, Troy, NY, USA, in 1990. He joined New Jersey Institute of Technology (NJIT), Newark, NJ, USA, in 1990, and has been a Distinguished Professor of Electrical and Computer Engineering since 2013. He has over 1400 publications including 17 books, over 900 journal papers (over 700 in IEEE Transactions), and 32 book-chapters. He holds 31 patents and several pending ones. His recently co-authored books include *Learning Automata and their Applications to Intelligent Systems*, IEEE Press/Wiley, Hoboken, NJ, 2024 (with J. Zhang), *Device-Edge-Cloud Continuum Paradigms, Architectures and Applications*, Springer Nature, 2023 (with C. Savaglio, G. Fortino, and J. Ma) and *Sustainable Manufacturing Systems: An Energy Perspective*, IEEE Press/Wiley, Hoboken, Hoboken, NJ, 2022 (with L. Li). His research interests are in intelligent automation, AI, Petri nets, robotics, Internet of Things, big data analytics, cloud/edge computing, transportation and energy systems.

Dr. Zhou is a recipient of the Excellence in Research Prize and the Medal from NJIT, the Humboldt Research Award for US Senior Scientists from Alexander von Humboldt Foundation, and the Franklin V. Taylor Memorial Award and the Norbert Wiener Award from IEEE SMC Society, the Computer-Integrated Manufacturing UNIVERSITY-LEAD Award from Society of Manufacturing Engineers, the Distinguished Service Award from IEEE Robotics and Automation Society, and the Edison Patent Award from the Research and Development Council of New Jersey. He served as an Editor-in-Chief for IEEE/CAA JOURNAL OF AUTOMATICA SINICA, an Associate Editor for IEEE TRANSACTIONS ON ROBOTICS AND AUTOMATION, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, and IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, and an Editor for IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING. He served as a Guest-Editor for many journals, including IEEE INTERNET OF THINGS JOURNAL, IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, and IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING. He is currently a Senior Editor of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, and an Associate Editor of Research, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE INTERNET OF THINGS JOURNAL, and *Frontiers of Information Technology & Electronic Engineering*. He is Founding Chair/Co-chair of Technical Committee on AI-based Smart Manufacturing Systems and Technical Committee on Humanized Crowd Computing of IEEE Systems, Man, and Cybernetics Society (SMC), Technical Committee on Semiconductor Manufacturing Automation of IEEE Robotics and Automation Society (RAS). He chaired IEEE North Jersey Section SMC Chapter from 2013 to 2023, and has been the Chair of the RAS Chapter since 2024. He is also a member of IEEE TAB Periodicals Committee and Periodicals Review and Advisory Committee. He served as the Vice-Chair for Fellow Evaluation Committee and the Vice-President for Conferences and Meetings of IEEE Systems, Man, and Cybernetics Society. He was a General Chair of IEEE Conference on Automation Science and Engineering, Washington D.C., August 23–26, 2008, a General Co-Chair of 2003 IEEE International Conference on System, Man and Cybernetics, Washington DC, October 5–8, 2003, 2019 IEEE International Conference on SMC, Bari, Italy, October 6–9, 2019, 2022 IEEE International Conference on SMC, Prague, Czech, October 9–12, 2022, and 2024 IEEE International Conference on SMC, Kuching, Malaysia, Oct. 6–10, 2024, the Founding General Co-Chair of 2004 IEEE International Conference on Networking, Sensing and Control, Taipei, March 21–23, 2004, and the General Chair of 2006 IEEE International Conference on Networking, Sensing and Control, Ft. Lauderdale, Florida, USA, April 23–25, 2006. He was the Program Chair of 2010 IEEE International Conference on Mechatronics and Automation, August 4–7, 2010, Xi'an, China, 1998 and 2001 IEEE International Conference on SMC and 1997 IEEE International Conference on Emerging Technologies and Factory Automation. He has led or participated in over 60 research and education projects with total budget over $ 14M, funded by National Science Foundation, Department of Defense, NIST, New Jersey Science and Technology Commission, and industry. He is a Life Member of Chinese Association for Science and Technology-USA and served as its President in 1999. He is Fellow of International Federation of Automatic Control, American Association for the Advancement of Science, Chinese Association of Automation, and National Academy of Inventors.

Authorized licensed use limited to: Khulna Univ of Engineering & Technology - KUET. Downloaded on February 16,2026 at 08:50:11 UTC from IEEE Xplore. Restrictions apply.